technology, generates an immune response. Although they effectively prevent severe disease and death, offering substantial protection against the virus, they are not 100 per cent effective at blocking infection from entering your body. However, in the case of an intra-nasal vaccine, it would target the nasal mucosal cells offering a big boost of immunity to your upper respiratory tract and the lungs. This would mean that not only would you not get infected since your upper respiratory tract would be protected, but you also wouldn't be able to infect others. Intra-nasal vaccines may offer significantly better protection also because they closely mimic the way the virus naturally infects humans, i.e., through the mucosal membranes of the nose and upper airways.

Mr Ella informed PM Modi that due to the limited case studies and requirements in the past, the testing protocols for such a vaccine weren't robust. He requested for support from the GoI and the MoHFW to expedite these procedural requirements, so as to speed up testing and development of their nasal vaccine candidate. (Six months on, by August 2021, they had already entered Phase II trials and were hopeful of robust data on the vaccines' safety and efficacy.)

The third component of their discussion was about production capacity constraints at their Genome Valley facility, which was capable of producing only about one crore doses per month. They informed the prime minister of their Bangalore facility, which could be repurposed to make about four to five crore doses per month and also requested for quick approvals and assistance so that production could be ramped up in time. In addition, they discussed with the prime minister a larger production augmentation plan which would include transferring technology and upgrading various public-sector manufacturers in order to achieve scale. With this information provided to the prime minister, it became clear that while they would be able to produce a large number of doses in 2021, the production would be pointedly back loaded wherein a disproportionately large number of doses would be produced in the latter half of the year. (Since then, a lot of action has taken place: inter-ministerial teams have visited them and more capacity-building exercises have been carried out where four more manufacturers were on boarded.) Within the next five months, ₹65 crore was provided by the GoI to Bharat Biotech to augment their Bangalore facility. Haffkine Bio-Pharmaceutical Corporation Ltd in Mumbai was provided ₹65 crore to build capacity to produce two crore doses a month